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Section B

System: Smart Parking Reservation System

Architecture Selection: Chose Client-Server Architecture for the Smart Parking Reservation System.

Reason 1 : Centralized, Consistent Reservation Data

The system must guarantee that a parking reservation validated at the gate camera is the same reservation record seen by the Reservation Manager and the Parking Warden's device. A client-server model keeps the reservation and license-plate records in a single authoritative Parking Database on the server, so gate cameras, the owner-facing reservation client, and the warden's handheld unit all read and write through the same server-side services. This avoids the data-consistency problems that a distributed microservices approach would introduce for a relatively small, single-facility deployment.

Reason 2 : Simple Deployment Matching a Single-Site Kiosk/Gate Setup

The parking facility has a fixed, bounded set of hardware clients (entry/exit gate cameras and warden handheld devices) that all communicate with one on-site or cloud-hosted server. Client-server keeps deployment and maintenance simple compared to microservices, which would add unnecessary operational overhead (service discovery, inter-service networking) for a system of this scale, and compared to a purely layered monolith, it cleanly separates hardware-facing clients from the server-side business logic and data store.

Security Advantage

Because all sensitive data — reservations, scanned license plates, and validation results — is stored and processed centrally in the Parking Database and Validation Service, access control, authentication of gate-camera and warden clients, and audit logging can be enforced at a single, well-defined server boundary rather than across many independently deployed services. This significantly reduces the attack surface compared to a microservices approach, where each service would need its own security perimeter.

Performance Benefit

Centralizing the Validation Service and Database on the server allows license-plate scan results from the Gate Camera to be validated against reservation records with a single, low-latency server-side lookup, rather than requiring multiple network hops between independently hosted

services (as microservices would require). This keeps gate processing fast enough for real-time vehicle entry, which is critical since drivers should not be delayed at the barrier.

Components Diagram:

